

ZIHAO LI (黎子豪)

✉ zihao.li@nbi.ku.dk  [0000-0001-5951-459X](https://orcid.org/0000-0001-5951-459X)  <https://lizihao1008.github.io/>

Education

University of Copenhagen

Ph.D. in Astronomy.

Advisor: Lise Christensen and Koki Kakiichi

Aug. 2024 –

Copenhagen, Denmark

Tsinghua University

M.Sc. in Astronomy. GPA: 3.85/4.00

Advisor: Zheng Cai

Sep. 2021 – Jun. 2024

Beijing, China

Sichuan University

B.Eng. in Aerospace Engineering. GPA: 3.80/4.00 rank: 2/28

Honors Degree in Top-notch Program.

Sep. 2017 – Jun. 2021

Chengdu, China

Research Interests & Experience

- Origin of the first stars.
- (Spatially resolved) metal enrichment of high- z galaxies.
- Environmental dependence of galaxy formation and evolution.
- Probe large scale structure through IGM tomography.

Honors & Awards

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| • 1 st Scholarship for Comprehensive Performance (10k CNY) Tsinghua University. | 2022 |
| • Award for Excellent Thesis Chinese National Level in Aeronautics & Astronautics. | 2021 |
| • MITACS Research Fellow (\$6k, cancelled due to COVID) University of Victoria. | 2020 |
| • 1 st Scholarship for Academic Performance Sichuan University. | 2020 |
| • Summer Abroad Subsidy (10k CNY) Sichuan University. | 2019 |
| • 1 st Scholarship of China Space Foundation (8k CNY) Sichuan University. | 2018 |
| • 1 st Scholarship for Comprehensive Performance (2.5k CNY) Sichuan University. | 2018 |

Publications

1st/2nd Author Papers:

- **Li, Z.**, Kakiichi, K., et al. Possible chemical signatures of first-star enrichment in a very metal-poor galaxy overdensity near the end of reionization. 2026, [Nature Astronomy](#)
- He, X. **Li, Z.**, et al. Revisiting the fundamental metallicity relation of dwarf galaxies at cosmic noon with NIRISS. 2026, [MNRAS, *stag1235*](#)
- **Li, Z.**, Cai, Z., et al. A 13-Billion-Year View of Galaxy Growth: Metallicity Gradient Evolution from the Local Universe to $z = 9$ with JWST and Archival Surveys. 2025, [ApJS](#)
- **Li, Z.**, Kakiichi, K., et al. Insights on Metal Enrichment and Environmental Effect at $z \approx 5 - 7$ with JWST ASPIRE/EIGER and Chemical Evolution Model. 2025, [A&A](#)
- **Li, Z.**, Cai, Z., et al. MAGNIF: A Tentative Lensed Rotating Disk at $z = 8.34$ detected by JWST NIRCам WFSS with Dynamical Forward Modeling. 2023, [arXiv:2310.09327](#)
- **Li, Z.**, Wang, X., Cai, Z., et al. First Census of Gas-phase Metallicity Gradients of Star-forming Galaxies in Overdense Environments at Cosmic Noon. 2022, [ApJL, 929, L8](#)
- Wang, X., **Li, Z.**, Cai, Z., et al. The Mass–Metallicity Relation at Cosmic Noon in Overdense Environments: First Results from the MAMMOTH–Grism HST Slitless Spectroscopic Survey. 2022, [ApJ, 926, 70](#)
- **Li, Z.**, Horowitz, B. and Cai, Z. Improved Ly α Tomography Using Optimized Reconstruction with Constraints on Absorption (ORCA). 2021, [ApJ, 916, 20](#)

Full list in [ADS](#).

Talks

- May. 2026. *Linking Metal-Poor Environments and Pop III Star Formation within a $z \sim 6$ Galaxy Overdensity*, contributed talk at STScI 2026 Spring Symposium. (Baltimore, US)
- June. 2025. *Environments and Metal Enrichment of Early Galaxies in Quasar Fields*, contributed talk at EAS 2025. (Cork, Ireland)
- May. 2025. *Metal enrichment and environmental effect in large scale structures at $z = 5 - 7$* , contributed talk at COSMOS 2025. (Marseille, France)
- May. 2025. *Environments and galaxy Formation from cosmic dawn to cosmic noon with slitless spectroscopy*, Cake talk at Technical University of Denmark. (Lyngby, Denmark)
- Jan. 2025. *JWST NIRCам WFSS data reduction workshop*, lecturer at NAOC. (Remote)
- Aug. 2024. *"Insights into cosmic evolution of gas-phase metallicity gradients with JWST"*, contributed talk at Santa Cruz Galaxy Workshop. (Santa Cruz, US)
- Dec. 2023. *Galaxy metal enrichment and gas dynamics with JWST*, seminar at NAOC. (Beijing, China)
- Nov. 2023. *"JWST insights on mass-metallicity relation and metallicity gradients of early galaxies"*, contributed talk at Nanjing University. (Nanjing, China)
- May. 2023. *"The metal-enrichment of low mass galaxies from cosmic dawn to noon in the JWST era"*, contributed talk at Chinese Astronomical Society Guoshoujing Symposium on Galaxies and Cosmology. (Huangshan, China)
- Mar. 2023. *"Research progress with HST/JWST slitless spectrograph and science preparation for CSST"*, contributed talk at China Space Station Telescope Conference 2023. (Huairou, China)

Teaching Experience

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| • TA of <i>Electrodynamics Lab</i> at University of Copenhagen. | Winter 2025 |
| • TA of <i>Electromagnetism Lab</i> at University of Copenhagen. | Fall 2025 |
| • TA of <i>Advanced Observational Astrophysics</i> at Tsinghua University. | Spring 2023 |

Professional Service

- Reviewer for ApJ, A&A, Nat. Commun.

Outreach Activities

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| • Founder of Astronomical Research Scholarship of Sichuan University (30,000 CNY). | 2025– |
| • Department Director of Sichuan University Astronomy Society. | 2018–2019 |
| • Co-Captain/Pilot/Technician of Sichuan University Students Aeromodel Team. | 2018–2019 |

References

Prof. Lise Christensen

✉ lichrist@nbi.ku.dk

Niels Bohr Institute, University of Copenhagen

Prof. Zheng Cai

✉ zcai@mail.tsinghua.edu.cn

Department of Astronomy, Tsinghua University

Prof. Xin Wang

✉ xwang@ucas.ac.cn

NAOC, Chinese Academy of Sciences

Dr. Benjamin Horowitz

✉ ben.horowitz@ipmu.jp

IPMU, Tokyo University